

PROJECT 3: PROBABILISTIC OPTIMIZATION Due date: May 22, 2026 (5:00pm PT)

In this project, you will perform multi-disciplinary optimization under uncertainty using a Gaussian Process (GP)—a probabilistic surrogate model representing a distribution over possible functions. Surrogate models are useful to predict the true objective function when it is costly, noisy, or challenging to evaluate directly. Gaussian Processes are particularly valuable because they quantify the uncertainty associated with their predictions. Your goal is to build a GP surrogate model representing the diversification efficiency of a simplified stock portfolio and use this model to determine the optimal number of stocks to invest in.

For projects, you may consult any material, including the textbook, online resources, classmates, and AI tools (e.g., ChatGPT), for guidance and advice only. All code must be written and typed up by you; do not copy and paste code from others or external sources.

General Instructions

Setup

You may use any programming language or package to fit and interact with the Gaussian Process. Popular GP libraries include `AbstractGPs.jl` in Julia and `sklearn.gaussian_process` in Python. Thoroughly review the documentation for your chosen package and ensure you have installed all necessary components before starting.

For Part 1, you will solve several constrained optimization problems. You may use external packages (e.g., `scipy.optimize` or `Optimization.jl`) or adapt your code from Project 2.

Problem Context

You manage an institutional equity fund, a pool of capital invested in stocks on behalf of large clients such as retirement funds, foundations, and university endowments. Your clients require that the fund hold stocks that are meaningfully different from the overall market, but not so extreme that the portfolio behaves like a completely different type of investment. These two requirements define the region in which you must construct your portfolio.

Each stock is characterized by two coordinates:

- **Market beta (β):** how sensitive the stock's returns are to broad market movements. A beta of 1 means the stock moves in lockstep with the market. For example, a high-risk technology company with $\beta = 1.75$ would return 175% of what the market returned in a given period. In contrast, a stable utility company with $\beta = 0.45$ would return only 45% of what the market returned.
- **High Minus Low (HML) loading:** a score measuring whether a stock is a *value* stock or a *growth* stock. High values indicate stocks that are cheap relative to what the company actually owns. For example, a large established manufacturer whose stock price is modest compared to its factories and equipment. Low values indicate stocks that are expensive relative to what the company actually owns. For example, a startup whose high stock price reflects what investors *expect* it to become, not what it currently has.

Together, these two coordinates place every stock as a point in a 2D grid. Your portfolio is therefore a set of n points in this grid. In this project you will:

1. Optimize the placement of n stocks in the 2D grid to maximize portfolio diversification.

2. Fit a Gaussian Process surrogate model to noisy observations of realized diversification efficiency.
3. Use the surrogate to predict expected profit and determine the optimal number of stocks to hold.

No prior knowledge of finance is required; all necessary information is provided.

Part 1: Portfolio Layout Optimization

Stocks with similar **Market beta** and **HML loading** respond to the same economic forces and tend to move together, providing little diversification benefit. In contrast, stocks with varying **Market beta** and **HML loading** respond to different forces and move more independently. As a result, simply adding more stocks does not automatically improve diversification; two nearly identical stocks contribute little more than one.

Your team has determined that the **diversification efficiency** η of a portfolio depends on the minimum L2 (Euclidean) norm distance p between any two holdings:

$$\eta = \frac{1}{1 + \frac{1}{p}}$$

As p increases, meaning holdings are more spread out in the 2D grid, efficiency approaches 1. As holdings cluster together, η falls toward 0. For a given number of stocks n , the goal is to find the layout that maximizes p and therefore maximizes η .

Feasible Region

Not all positions in the 2D grid are permitted. Your fund's investment mandate imposes two types of restrictions that define a feasible region, which is neither too close to the benchmark nor too far from it.

The two mandate constraints are:

- **Active share minimum (inner boundary).** Your clients require the fund's portfolio to be sufficiently different from its benchmark, the S&P 500 index, which tracks the performance of 500 of the largest publicly traded companies in the United States. Active share measures how different your holdings are from the benchmark; a higher active share means a more distinctive portfolio. In the 2D grid, this creates a circular forbidden zone centered on the benchmark at (5.0, 5.0) with radius $r = 1.5$. No stock may be placed within this inner circle.
- **Tracking error ceiling (outer boundary).** Your clients also require that the portfolio not deviate so far from the S&P 500 index that it loses its character as an equity fund. Tracking error is a measure of how closely an investment portfolio follows the performance of a benchmark index. Therefore, stocks placed too far from the benchmark in the 2D grid may cause the fund to perform vastly differently. This constraint creates a maximum allowable distance of $R = 4$ from the benchmark center. No stock may be placed outside this outer circle.

Together, these two constraints confine every stock to the ring between radius $r = 1.5$ and radius $R = 4.0$ from the benchmark.

Two additional sub-regions within this feasible region are also excluded because certain specific combinations of **beta**, a stock's volatility, and **HML**, where the stock sits on the value-to-growth spectrum, can cause the fund's performance to deviate from the benchmark:

- **High-beta growth zone** (center (7.5, 2.5), radius 1.2): Stocks in this region have high market sensitivity and a strong growth tilt. A single position here amplifies systematic market risk enough to violate the tracking error.

- **Defensive deep value zone** (center (2.5, 7.5), radius 1.2): Stocks in this region have low market sensitivity and an extreme value tilt. These stocks diverge from the benchmark along the value-growth dimension, which can violate the tracking error through the HML channel alone.

The resulting feasible and infeasible regions are shown in Figure 1. All four constraints are summarized in Table 1.

Table 1: Feasible region constraints normalized in the 2D grid. The benchmark is located at (5.0, 5.0).

Constraint	Type	Parameter	Mandate source
Inner boundary — benchmark core	Must be <i>outside</i>	$r = 1.5$ from (5.0, 5.0)	Active share minimum
Outer boundary — tracking error ceiling	Must be <i>inside</i>	$R = 4.0$ from (5.0, 5.0)	Tracking error ceiling
High- β growth zone	Must be <i>outside</i>	$r = 1.2$ from (7.5, 2.5)	Tracking error: β channel
Defensive deep value zone	Must be <i>outside</i>	$r = 1.2$ from (2.5, 7.5)	Tracking error: HML channel

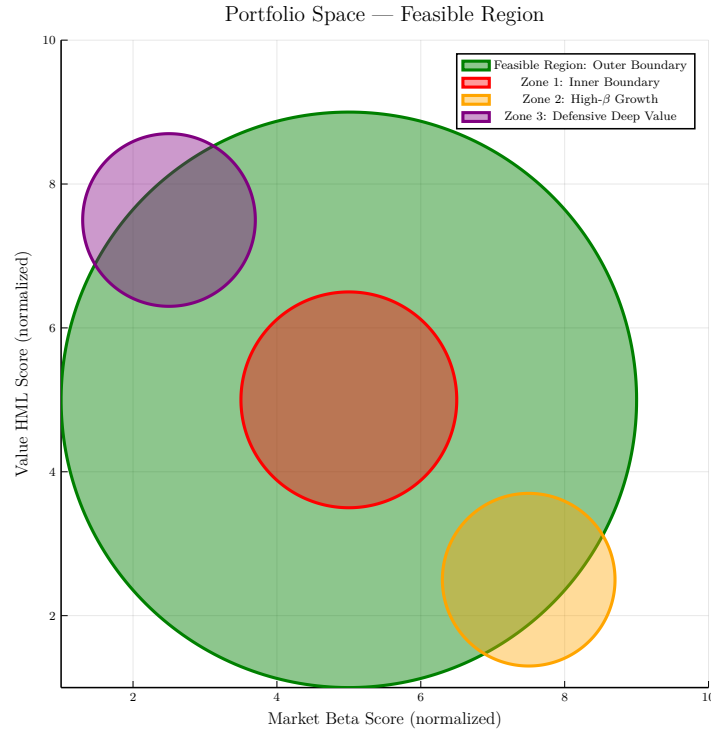


Figure 1: Feasible region normalized in the 2D grid. The green annulus is the region where stocks may be placed: outside the inner circle (active share constraint), inside the outer circle (tracking error ceiling), and outside the two red sector exclusions.

Your Tasks

1. Solve the optimization problem for portfolio sizes $n \in [2, 10]$ where $n \in \mathbb{Z}$ (i.e., solve the problem for 9 different portfolio sizes, 2 through 10 stocks). Formally, the objective is to maximize the minimum pairwise L2 norm distance over all stock positions. Let p^* denote the optimal minimum pairwise distance achieved for a given n :

$$p^* = \max_{\{s_i\}} \min_{i \neq j} \|s_i - s_j\|_2,$$

subject to all the feasibility constraints above. You should expect p^* to decrease as n grows; fitting more points into the same feasible region forces some of them closer together.

Note: This optimization problem becomes harder as n grows. We recommend using some mix of random restarts, multiple methods, and stochasticity to find the optimum for each portfolio size.

2. Output your solutions for all configurations ($n \in [2, 10]$) as a comma-separated values (CSV) file and submit your solutions to the Autograder on Gradescope. The CSV file should not have headers. The first column is the number of stocks, and the second column is the optimized minimum pairwise distance p^* (i.e., the objective value achieved by your best layout) for that configuration. The filename can be anything you want, but it must have a `.csv` extension.

Example of a correctly formatted submission file:

```
2,1.2345
3,12.3456
4,123.4567
5,1234.5678
6,0.1234
7,123.4567
8,123.4567
9,123.4567
10,123.4567
```

3. Plot the resulting stock layout for the following cases. Include all constraint boundaries in each plot.
 - (a) $n = 3$ stocks
 - (b) $n = 5$ stocks
 - (c) $n = 7$ stocks
4. Plot the minimum pairwise distance p^* (the values in your CSV file) against the number of stocks for all configurations $n \in [2, 10]$. Plot the minimum distance on the y-axis against the number of stock positions on the x-axis.
5. Calculate the theoretical diversification efficiency for your solutions using

$$\eta = \frac{1}{1 + \frac{1}{p^*}}$$

where p^* is the minimum pairwise distance from Task 2 (the values in your CSV file). Plot efficiency on the y -axis against minimum separation distance on the x -axis.

Part 2: Gaussian Process Fitting

Based on prior work, the team knows that the theoretical diversification efficiency does not always match the efficiency realized in the stock market. The team therefore conducted high-fidelity simulations to obtain more accurate efficiency estimates at specific separation distances. The simulation results are noisy with a **standard deviation of 0.02**.

Here is the data from the simulations:

Min. separation	Measured efficiency
2.5	0.59
2.5	0.62
4.5	0.82
4.5	0.85
6.5	0.86
6.5	0.88
8.0	0.90
8.0	0.91

Your Tasks

1. Fit a Gaussian Process to the simulation data above using the **squared exponential kernel with length scale $\ell = 4.0$** and the **theoretical efficiency formula**

$$\eta(p) = \frac{1}{1 + \frac{1}{p}}$$

as the GP mean function, so that the model is anchored to financial theory in regions with no data.

Note: Ensure that your package does not automatically optimize the hyperparameters.

Hint: For guidance on how to plot GPs, check out the tutorial videos on the *project 3* page.

2. Produce a plot in the style of Figure 18.7 (second edition) of the textbook where the uncertainty bounds represent the 95% confidence interval ($\pm 1.96\sigma$). Plot your GP over the range $p \in [1, 10]$ to show how uncertainty grows away from the observed data. The plot should include:
 - (a) Separation distance on the x -axis and efficiency on the y -axis.
 - (b) The predicted GP mean function.
 - (c) The 95% confidence region.
 - (d) The simulation data points used to fit the GP.
 - (e) The theoretical efficiency curve plotted on the same axes.

Part 3: Determining the Optimal Portfolio Size

The team now wants to know how many stocks to hold in order to maximize expected portfolio profit. The expected yearly profit from an n -stock portfolio is:

$$P(n) = \underbrace{n \times \eta_n \times \text{capital}_{\text{stock}} \times \alpha_{\text{stock}}}_{\text{total revenue}} - \underbrace{n \times c_{\text{stock}}}_{\text{total cost}}$$

where:

η_n = diversification efficiency of a portfolio of n stocks

$\text{capital}_{\text{stock}}$ = capital allocated per stock position = \$25,000,000

α_{stock} = expected annualized return above what the market delivers per position = 4%

c_{stock} = annual cost per stock position = \$550,000

Note: The gross alpha per stock reflects the expected active return, the number of trading days, and the expected fraction of days the position is active.

Your Tasks

1. Use your GP surrogate to calculate the mean profit with a 95% confidence region across portfolio sizes $n \in [2, 10]$. You will produce a figure of expected profit vs. number of stocks n , analogous to Figure 18.7 in the textbook, to support your analysis. Consider only integer values of n .
2. Provide a brief justification of your recommended portfolio size by addressing the following points:
 - (a) How many stocks would you recommend if you ignored uncertainty and used only the theoretical efficiency?
 - (b) How many stocks would you hold to maximize *expected* profit (GP mean)?
 - (c) Does your recommendation change if you want to maximize *worst-case* profit, defined as the lower bound of the 95% confidence interval?
 - (d) How many stocks would you hold to maximize *best-case* profit, defined as the upper bound of the 95% confidence interval?
 - (e) Give a reason why a portfolio manager might design to the mean, worst-case, or best-case rather than the others.
 - (f) If the team were to conduct additional simulations, at what separation distances would you recommend collecting new data? Justify your answer in terms of how new observations would reduce uncertainty and potentially change the optimal portfolio size recommendation.

Submission

In addition to the Autograder submission for Part 1, please submit a PDF containing the following:

- **Part 1: Portfolio layout optimization**

- Scatter plot for $n = 3$ stocks
- Scatter plot for $n = 5$ stocks
- Scatter plot for $n = 7$ stocks
- Plot of minimum pairwise distance p^* vs. number of stocks for all $n \in [2, 10]$
- Plot of minimum separation distance vs. theoretical diversification efficiency

- **Part 2: Gaussian Process fitting** (refer to Fig. 18.7, 2nd edition)

- Plot of the GP fit to the simulation data. The GP fit plot must contain the predicted mean, 95% confidence region, simulation data, and theoretical efficiency curve. Remember to use the squared exponential kernel with $\ell = 4.0$ and the theoretical efficiency as the mean function.

- **Part 3: Optimization under uncertainty**

- Figure of expected profit vs. number of stocks
- Brief discussion addressing points 1–6 in Part 3

- **Your code.**